

From Raw Sanger Chromatograms to Phylogenetic Analysis: A Beginner Bioinformatics Workflow Using Fungal ITS Sequences

Project Overview

This project demonstrates an end-to-end beginner bioinformatics workflow using raw Sanger sequencing chromatogram data obtained from a publicly available Zenodo dataset. The workflow includes chromatogram quality assessment, sequence trimming, sequence alignment, BLAST identification, and phylogenetic analysis using fungal ITS sequences.

Objective

The objective of this project was to understand how raw sequencing data are processed into biologically interpretable results using standard bioinformatics tools and workflows.

Workflow Summary

- Selection of raw .ab1 chromatogram files from Zenodo dataset
- Chromatogram quality assessment in Chromas
- Forward and reverse read alignment in MEGA12 using MUSCLE
- Manual trimming of low-quality and non-overlapping regions
- Generation of consensus sequence
- NCBI BLAST analysis
- Retrieval of top matching ITS reference sequences
- Multiple sequence alignment
- Neighbor-Joining phylogenetic tree construction with bootstrap analysis

Chromatogram Quality Assessment Screenshot

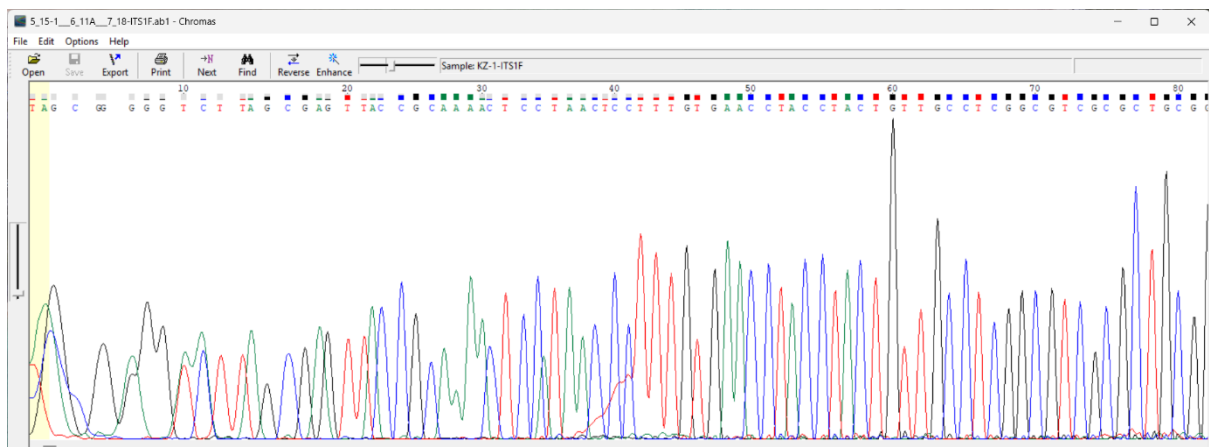


Figure 1: Chromatogram showing poor quality read in the beginning of a forward sequence

BLAST® >> blastn suite >> results for RID-2C64SSYC014

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Job Title

Sample 1

RID

2C64SSYC014 Search expires on 06-09 09:19 am [Download All ▼](#)

Program

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Database

core_nt [See details ▼](#)

Query ID

lcl|Query_1741251

Description

KZ-1-ITS1R

Molecule type

dna

Query Length

202

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	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
✓	Annulohypoxyylon sp. isolate CSC21B0684 small subunit ribosomal RNA gene, partial sequence: internal transcrib...	Annulohypoxyylon...	374	374	100%	4e-99	100.00%	619	PQ821603.1
✓	Annulohypoxyylon annulatum isolate N32 18S ribosomal RNA gene, partial sequence: internal transcribed spacer...	Annulohypoxyylon...	374	374	100%	4e-99	100.00%	640	FJ481156.1
✓	Annulohypoxyylon annulatum isolate N29 18S ribosomal RNA gene, partial sequence: internal transcribed spacer...	Annulohypoxyylon...	374	374	100%	4e-99	100.00%	604	FJ481155.1
✓	Annulohypoxyylon sp. CF_K220_1 genes for 18S rRNA, ITS1, 5.8S rRNA, partial and complete sequence	Annulohypoxyylon...	374	374	100%	4e-99	100.00%	279	LC750340.1
✓	Annulohypoxyylon annulatum genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequ...	Annulohypoxyylon...	374	374	100%	4e-99	100.00%	621	LC317411.1
✓	Annulohypoxyylon sp. isolate JODH102-2 internal transcribed spacer 1, partial sequence: 5.8S ribosomal RNA gen...	Annulohypoxyylon...	374	374	100%	4e-99	100.00%	599	ON557328.1
✓	Annulohypoxyylon sp. HMF-082 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete seq...	Annulohypoxyylon...	374	374	100%	4e-99	100.00%	617	LC428833.1
✓	Annulohypoxyylon annulatum voucher 14143 small subunit ribosomal RNA gene, partial sequence: internal transcri...	Annulohypoxyylon...	374	374	100%	4e-99	100.00%	611	MW023097.1
✓	Fungal sp. isolate ARIZ-Torreya-BES041 internal transcribed spacer 1, partial sequence: 5.8S ribosomal RNA gen...	fungal sp.	374	374	100%	4e-99	100.00%	1071	PQ576794.1
✓	Annulohypoxyylon annulatum strain MO313254 small subunit ribosomal RNA gene, partial sequence: internal trans...	Annulohypoxyylon...	374	374	100%	4e-99	100.00%	659	MH558286.1
✓	Fungal endophyte strain SV1857 18S ribosomal RNA gene, partial sequence: internal transcribed spacer 1, 5.8S...	fungal endophyte	374	374	100%	4e-99	100.00%	1176	KY765270.1
✓	Annulohypoxyylon annulatum isolate AH358 small subunit ribosomal RNA gene, partial sequence: internal transcri...	Annulohypoxyylon...	374	374	100%	4e-99	100.00%	627	PP357844.1

Phylogenetic Tree Screenshot

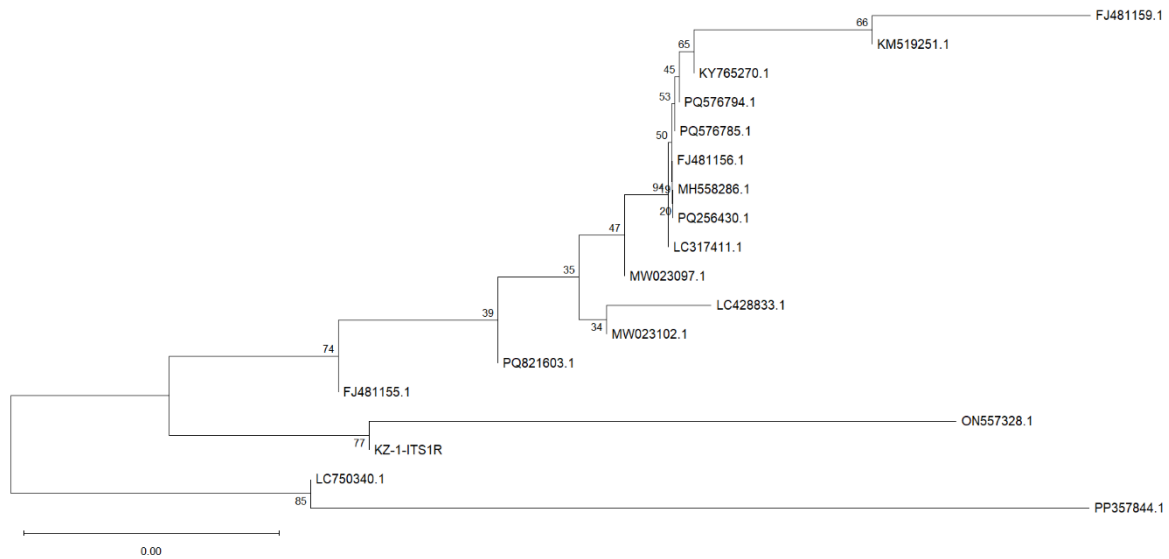


Figure 3: Phylogenetic Tree produced using MEGA12 showing the relationship of sample with other species

Results and Interpretation

BLAST analysis showed strong similarity to *Annulohypoxylon*-associated fungal ITS sequences with 100% query coverage and identity among top hits. Phylogenetic analysis placed the sample within an *Annulohypoxylon*-related clade and demonstrated evolutionary relatedness with selected reference sequences.

Tools Used

Tools	Purpose
Chromas	Chromatogram visualization and quality assessment
MEGA12	Sequence alignment and phylogenetic analysis
MUSCLE	Multiple sequence alignment
NCBI BLAST	Sequence similarity search and identification

Conclusion

This project successfully demonstrated a complete beginner-level Sanger sequencing bioinformatics workflow, beginning with raw chromatogram data and ending with phylogenetic interpretation. The project provided practical experience in sequence quality control, sequence alignment, BLAST analysis, and evolutionary analysis using phylogenetic trees.